Hydraulic Elevator Maintenance Manual

Rocket Elevators - Ontario Fleet Technical Reference

Overview

Hydraulic elevators use a fluid-driven piston to raise and lower the cab. Routine maintenance focuses on fluid integrity, pump condition, cylinder seals, and control valve operation. TSSA requires a full hydraulic test every five years and annual oil sampling for in-ground cylinders.

Hydraulic Fluid - Inspection and Change Intervals

Check fluid level monthly. The fluid should be clear to amber with no milky discolouration (which indicates water ingress). Sample the fluid annually; test for viscosity, acid number, and water content per ASTM D4378. Replace fluid when viscosity exceeds 20% of original grade or acid number exceeds 2.0 mg KOH/g. Dispose of used hydraulic oil as Class 3A hazardous waste - never drain into sewers.

Hydraulic Power Unit - Pump and Motor

Inspect the pump coupling and motor mounts quarterly. Listen for cavitation (high-pitched whine at start of UP travel) - this indicates low fluid or a clogged suction filter. Replace the suction strainer annually. Check motor amperage draw against nameplate: sustained over-amperage accelerates winding failure. Re-grease motor bearings every 2,000 operating hours with NLGI Grade 2 lithium-complex grease.

Control Valve - Adjustment and Testing

The control valve governs up-direction speed, down-direction speed, and pressure relief. Adjust lowering speed (down valve) so the car decelerates smoothly into each landing - typical target is 0.1 m/s at final approach. The pressure relief valve must be set to no more than 125% of working pressure per ASME A17.1 Section 3.19. Test annually by closing the main stop valve and running the pump: relief must open cleanly without chatter.

Cylinder and Piston Seal Inspection

For above-ground cylinders, inspect the piston rod seal annually for weeping. A slow seep (< 1 drop / 10 minutes) is acceptable; a drip requires seal replacement within 30 days. For single-bottom in-ground cylinders, conduct an oil sampling program per CAN/CSA B44 and monitor for groundwater contamination. If a cylinder failure is suspected, shut the elevator down and notify TSSA before any excavation begins.

Pressure Test Procedure

1. Load the car to 125% of rated capacity.
2. Close the main stop valve.
3. Run the pump to achieve 125% of working pressure.
4. Hold for five minutes - pressure must not drop more than 5%.
5. Open main stop valve; lower car at rated speed, check levelling.
6. Record results on TSSA Form EL-HYD-01.

Common Hydraulic Faults and Remedies

Slow down travel: Check fluid viscosity, inspect suction filter, verify motor voltage.
Car creeps down (levelling drift): Adjust or replace down valve, check for seat wear.

Rough ride: Air in system - bleed via bleeder screw at top of cylinder.
Oil on pit floor: Inspect power unit connections, cylinder base seal, return-line fittings.